ASSIGNMENT-7

NAME: TANMOY BANIK

ROLL:CSE 206

Data: { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },

{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },

{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },

{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },

{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },

{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },

{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },

{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }.

Q1. Display all documents

Query: db.studentCSE206.find()

Output:

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	1	Anu	22	IT	90	Pass	A
69c81d5c896d12...	2	Riya	21	CSE	78	Pass	C
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A
69c81d5c896d12...	4	Neha	20	ECE	74	Pass	C
69c81d5c896d12...	5	Raj	24	CSE	88	Pass	B
69c81d5c896d12...	6	Simran	22	IT	81	Pass	B
69c81d5c896d12...	8	Priya	23	CSE	95	Pass	A
69c81f6e896d12a...	9	Arjun	22	IT	87	Pass	B

Q2. Find students from IT department

Query: db.studentCSE206.find({ department: "IT" })

Output:

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	1	Anu	22	IT	90	Pass	A
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A
69c81d5c896d12...	6	Simran	22	IT	81	Pass	B
69c81f6e896d12a...	9	Arjun	22	IT	87	Pass	B

Q3. Find students with marks > 80

Query: db.studentCSE206.find({ marks: { $gt: 80 }})

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	1	Anu	22	IT	90	Pass	A
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A
69c81d5c896d12...	5	Raj	24	CSE	88	Pass	B
69c81d5c896d12...	6	Simran	22	IT	81	Pass	B
69c81d5c896d12...	8	Priya	23	CSE	95	Pass	A
69c81f6e896d12a...	9	Arjun	22	IT	87	Pass	B

Q4. Insert one new student

Query: db.studentCSE206.insertOne({ roll: 9, name: "Arjun", age: 22, department: "IT", marks: 87 })

Output:

acknowledged	insertedId	matchedCount	modifiedCount	upsertedCount
true	null	1.0	0.0	0.0

Q5. Update marks of Anu to 90

Query: db.studentCSE206.updateOne**({** name**:** "Anu" **},** **{** $set**: {** marks**:** 90 **}})**

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q6. Add a new field "status: Pass" for all students

Query: db.studentCSE206.updateMany({},{ $set: { status: "Pass" }})

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q7. Delete all students with marks < 70

Query: db.studentCSE206.deleteMany**({** marks**: {** $lt**:** 70 **} })**

Output:{ acknowledged: true, deletedCount: 0 }

Q8. Count students in CSE department

Query: db.studentCSE206.countDocuments({ department: "CSE" })

Output:3

Q9. Find total students in each department

Query: db.studentCSE206.aggregate([{ $group: { _id: "$department", totalStudents: { $sum: 1 }}}])

Output:

_id	totalStudents
CSE	3.0
IT	4.0
ECE	1.0

Q10. Find highest marks

Query: db.studentCSE206.find().sort({ marks: -1 }).limit(1)

Output:

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A

Q11. Find the second highest marks

Query: db.studentCSE206.find().sort({ marks: -1 }).skip(1).limit(1)

Output:

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A

Q12. Find students whose name starts with 'A' and marks > 80

Query: **db.studentCSE206.find({**

name: { $regex: "^A" },

marks: { $gt: 80 }})

Output:

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	1	Anu	22	IT	90	Pass	A
69c81d5c896d12...	3	Aman	23	IT	92	Pass	A
69c81f6e896d12a...	9	Arjun	22	IT	87	Pass	B

Q13. Display department-wise average marks, only for departments where avg > 80

Query: db.studentCSE206.aggregate([{ $group: { _id: "$department", avgMarks: { $avg: "$marks" } } }, { $match: { avgMarks: { $gt: 80 } } }])

Output:

_id	avgMarks
CSE	87.0
IT	87.5

Q14. Find students with marks between 75 and 90, sorted by marks ascending

Query: db.studentCSE206.find({ marks: { $gte: 75, $lte: 90 }}).sort({ marks: 1 })

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 0,
  upsertedCount: 0
}
```

Q15. Add grade field based on marks

Query: db.studentCSE206.updateMany({}, [{ $set: { grade: { $switch: { branches: [{ case: { $gte: ["$marks", 90] }, then: "A" }, { case: { $gte: ["$marks", 80] }, then: "B" }, { case: { $gte: ["$marks", 70] }, then: "C" }], default: "D" } } } }])

Output:

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 0,
  upsertedCount: 0
}
```

_id	roll	name	age	department	marks	status	grade
69c81d5c896d12...	1	Anu	22	IT	90	Pass	A
69c81d5c896d12...	2	Riya	21	CSE	78	Pass	C
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69c81d5c896d12...	4	Neha	20	ECE	74	Pass	C
69c81d5c896d12...	5	Raj	24	CSE	88	Pass	B
69c81d5c896d12...	6	Simran	22	IT	81	Pass	B
69c81d5c896d12...	8	Priya	23	CSE	95	Pass	A
69c81f6e896d12a...	9	Arjun	22	IT	87	Pass	B